

1. Research Questions

Topic: Biological Sex vs. Height in U.S. High School Students.

Research Question: Is there a significant difference in mean height between males and females?

2. Group Definition

Dataset: CDC 2007 YRBS (N = 13,062)

Female(Baseline): $n_1 = 6,490$

Male(Comparison): $n_2 = 6,572$

3. Response Variable

Variable: Height (Meters/Continuous)

Data Cleaning: Retained [0.5m – 2.5m];

Excluded missing (NaN) & invalid code

4. Method

Test: Welch's Two-Sample t-test($\alpha=0.05$)

Assumption: Equal variances NOT assumed

Hypotheses: $H_0: \mu_{\text{male}} = \mu_{\text{female}}$ vs. $H_a: \mu_{\text{male}} \neq \mu_{\text{female}}$

Justification: Accounts for unequal variances, large N satisfies Central Limit Theorem (CLT).

5. Key Result

Welch's t-test Statistics:

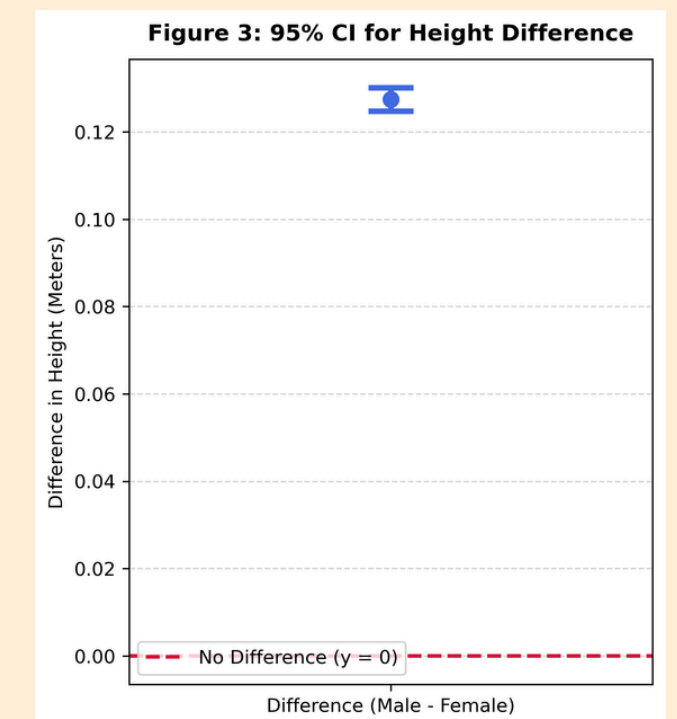
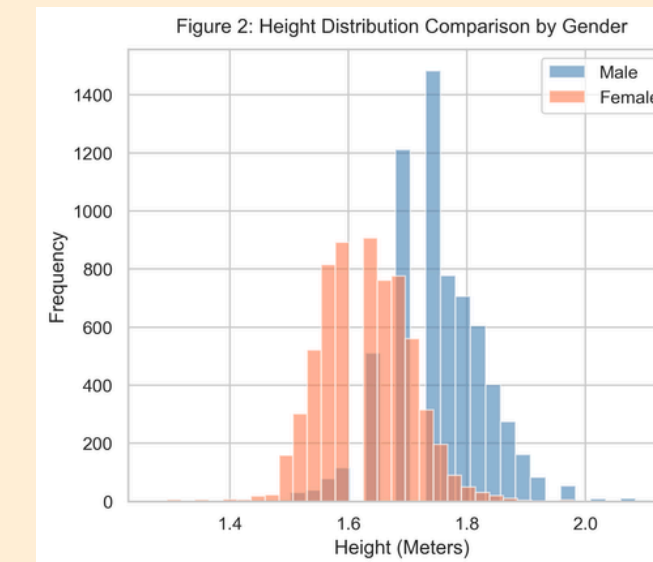
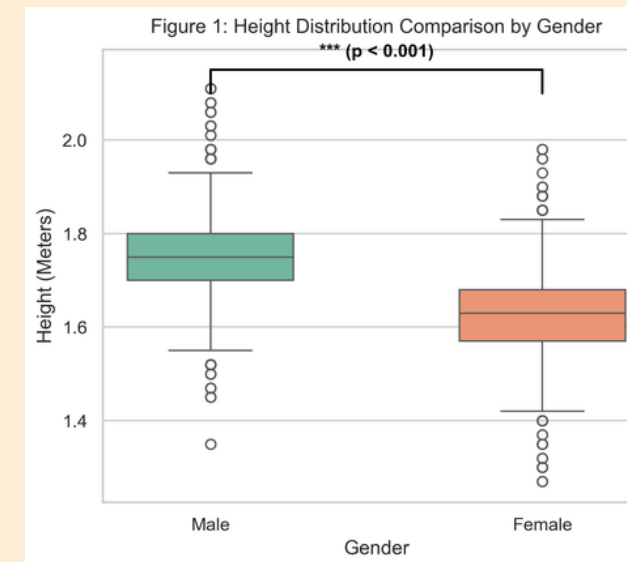
$t = 92.31$

$p < 0.001$

Effect Estimate & Confidence Interval:

Mean Difference: 0.1275 m

95% Confidence Interval: [0.1247, 0.1302] m



6. Conclusion

Decision: Reject H_0 ($p < 0.001$)

Finding: U.S. male high school students ($\mu_{\text{male}} = 1.7574$ m) are significantly taller than female students ($\mu_{\text{female}} = 1.6299$ m) by an average of 12.75 cm